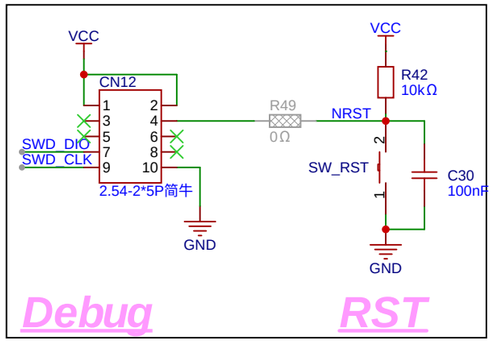
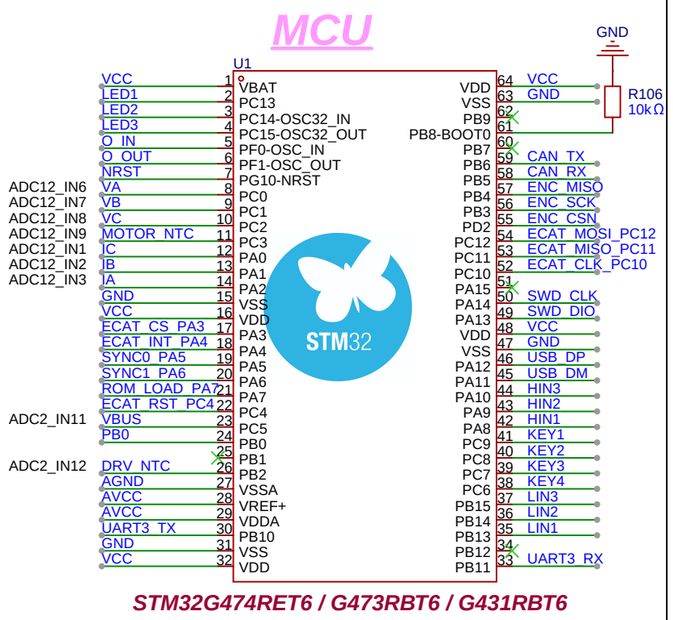
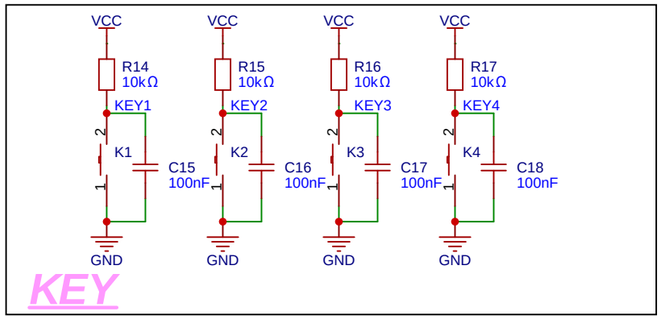
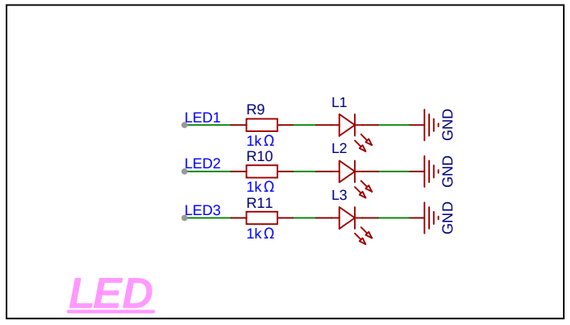
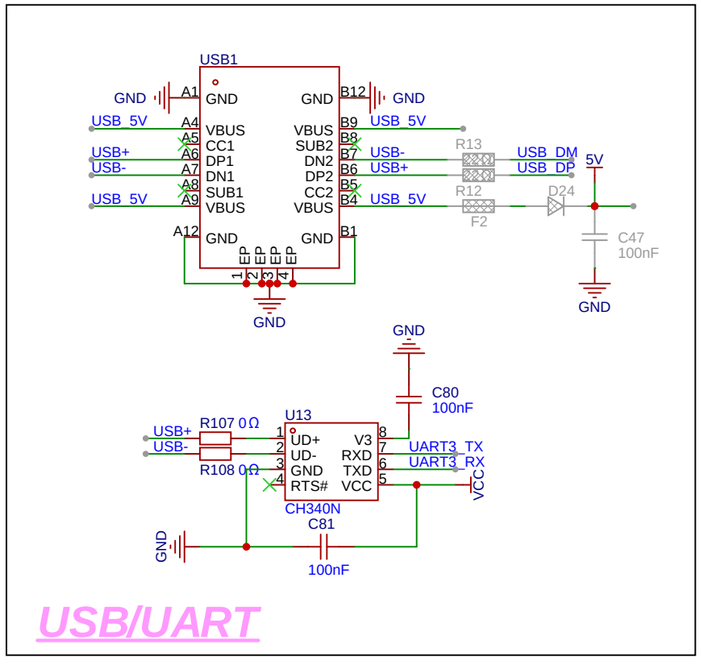


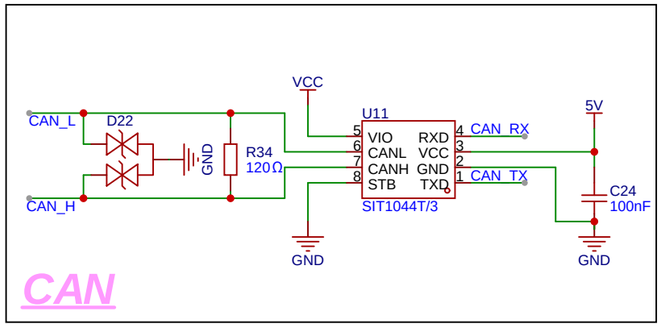


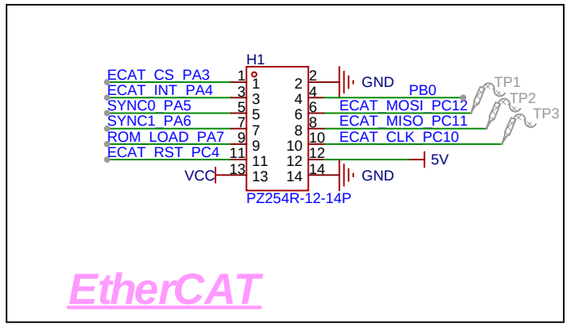




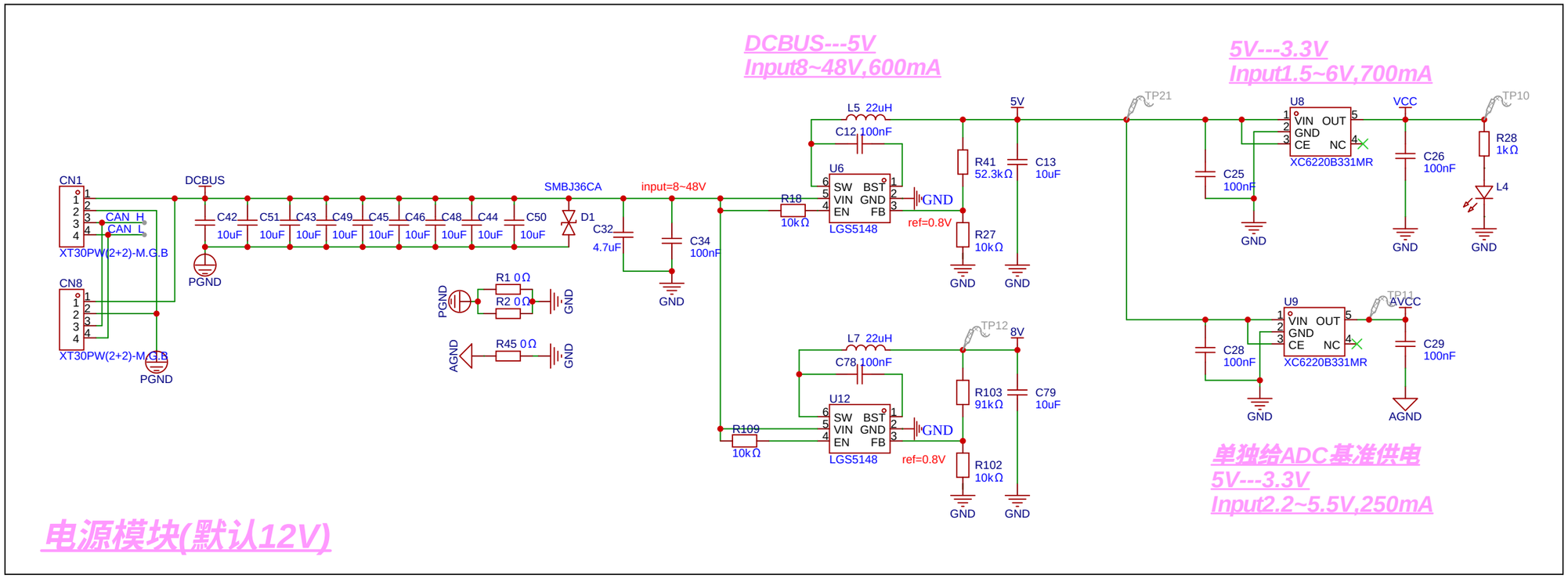




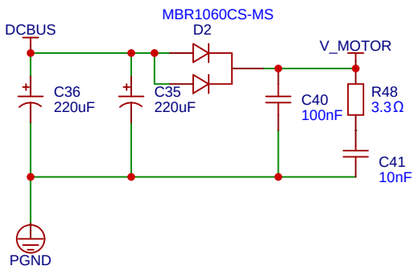




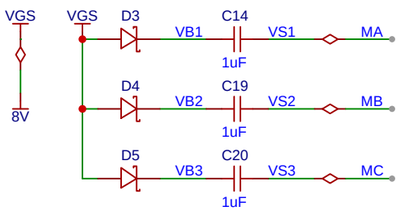
TITLE: iRDT_Driver		REV: V1.4
机器人驱动小部落	Company: 机器人驱动小部落	Sheet: 1/3
	Date: 2026-07-30 Drawn By: intelligent Robot-Driven Tribe	



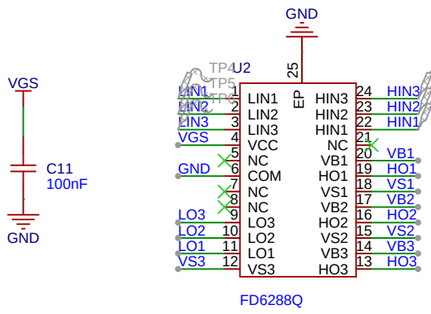
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	Date: 2026-07-30 Drawn By: intelligent Robot-Driven Tribe	



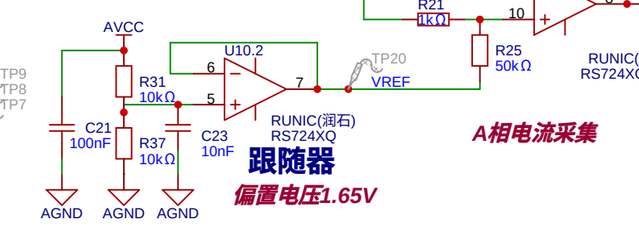
电源处理



自举电路

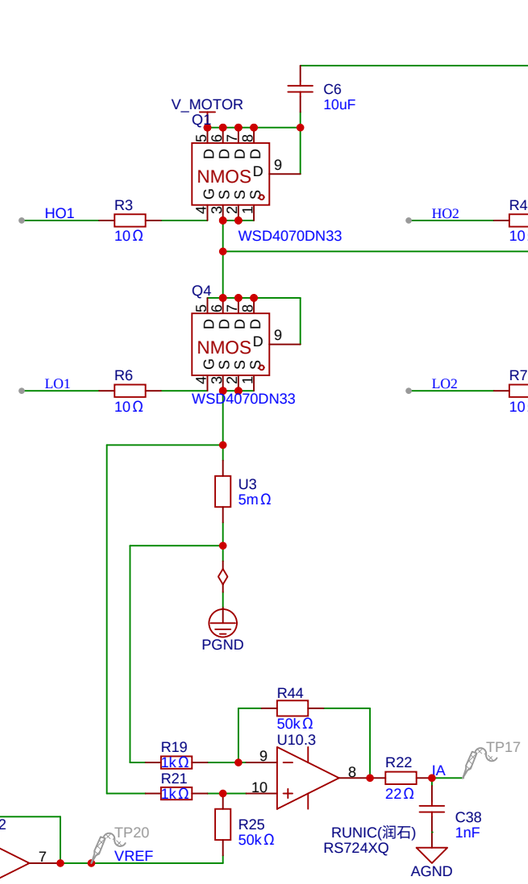


预驱芯片



跟随器

偏置电压1.65V



正面贴片，用于小电机

MOS驱动电路

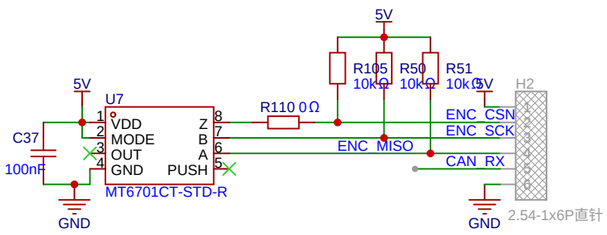
背面焊接，用于大电机

A相电流采集

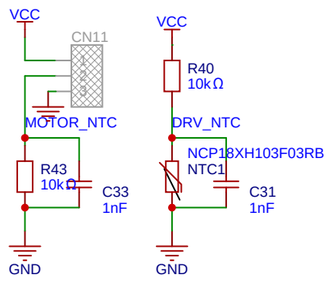
B相电流采集

C相电流采集

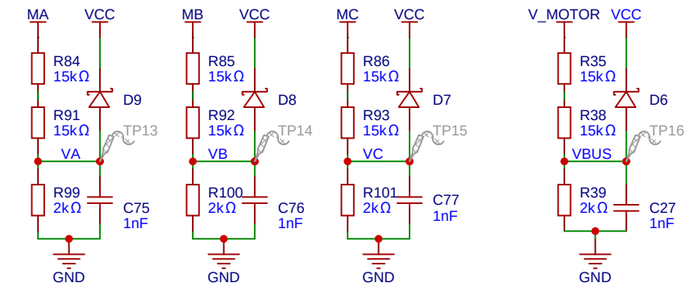
电流采样



MT6701CT编码器



温度采样



A相电压采样

B相电压采样

C相电压采样

母线电压采样

TITLE: iRDT_Driver		REV: V1.4
机器人驱动小部落	Company: 机器人驱动小部落	Sheet: 3/3
	Date: 2026-07-30 Drawn By: intelligent Robot-Driven Tribe	

采样点最高电压为: 3.3V + 二极管管压降

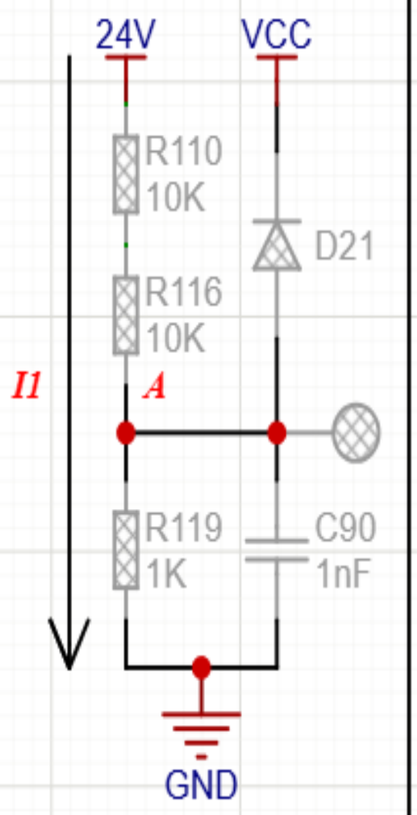
采样点电压应接近3.3V的一半,分辨率高，易采样

I1电流: 24V/(10k+10k+1k) = 1.14mA

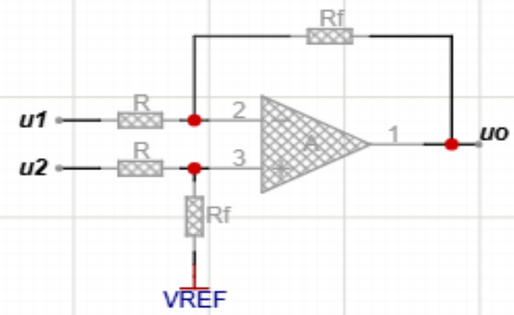
A点电压: I1 * 1k = 1.14V

缩小电压系数: k = (10k+10k+1k) / 1k

VA = adc_val * (3.3V/4096) * k



电流采样计算



运放增益:k = Rf / R

偏置电压:VREF = 1.65V = 2048

采样电阻:Rm = 1mΩ

运放输出:uo = k * (u2 - u1)

u2 - u1 = I * Rm

uo = (adc_val - 2048) * 3.3V/4096

采集电流:I = uo / (Rm * k)